

Nishan Kharel

AI Engineer & Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with **2+ years of hands-on experience** developing machine learning, NLP, computer vision, and Generative AI solutions. Proven track record delivering end-to-end AI pipelines from data preprocessing to production deployment using Python, SQL, LLMs, RAG, and deep learning frameworks. Skilled at translating complex data challenges into scalable AI solutions for recruitment, document intelligence, and real-world applications.

EDUCATION

Bachelor of Computer Science and Information Technology 2026 Feb, Completed with Aggregate 80.5 Percentage
Texas International College, Tribhuvan University – Kathmandu, Nepal

TECHNICAL SKILLS

Languages	Python, C, SQL
Machine Learning & AI	Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP), Computer Vision
LLMs & Fine-tuning	Large Language Models (LLMs), Fine-tuning, QLoRA, LoRA, PEFT, Transformers, Prompt Engineering, Hugging Face
RAG & Vector Search	Retrieval-Augmented Generation (RAG), pgvector, Vector Databases, Embeddings, Semantic Search
Frameworks & Libraries	PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, Jupyter
Backend & APIs	FastAPI, Flask, REST APIs, SQLAlchemy (ORM), Alembic, SSL/TLS Configuration
Cloud & MLOps	AWS, Amazon SageMaker, AWS RDS, GPU Training & Acceleration, Remote Server Deployment
Databases	PostgreSQL, pgvector, pgAdmin, Relational Databases
DevOps & Dev Tools	Docker, Docker Hub, Git, Postman, Swagger, TablePlus, VS Code

PROFESSIONAL EXPERIENCE

Applied Scientist

Samni Labs AI · USA, Remote

Feb-05, 2026

- Fine-tuned Transformer based Models, Healthcare data Processing and AI automation.

AI Engineer (Trainee/Internship)

Mantra Ideas Tech · Kathmandu, Nepal

Oct 2025 – Feb 2026

- Engineered an **AI-based Bank Note OCR System** using computer vision and deep learning to detect and classify currency denominations, improving recognition accuracy and operational reliability.
- Fine-tuned **Text-to-Speech (TTS) models** for Nepali voice synthesis, optimising speaker adaptation and naturalness for real-time speech generation with VibeVoice and Voice Cloning.

Data Scientist (Internship + Full Time)

Mind Risers Technology · Kathmandu, Nepal

Jan 2025 – Sep 2025

- Developed an **AI-powered ATS system** for Recruit Nepal using Tesseract OCR, spaCy NER, and semantic search to match candidate profiles with job descriptions, improving recruiter efficiency and user engagement.
- Built a **Nepali fake news classifier** for the Pop-Up app using BERT-based Transformer models, enabling real-time high-accuracy content verification.
- Created an **automated Job Description Generator** with secure API integration, enabling dynamic job-posting creation from role and organisational inputs.

Data Science Consultant (Part-time)

Namaste Network · Chabahil, Kathmandu

Feb 2025 – Feb 2026

- Designed and deployed end-to-end data science solutions covering predictive modelling, NLP pipelines, and automated analytics using Python and SQL.
- Delivered training on Neural Networks and Transformer architectures, covering backpropagation, attention mechanisms,

and real-world NLP applications.

- Designed data science bootcamp curriculum and materials, **improving student satisfaction by 25%**.

Machine Learning Tutor

Texas International College · Kathmandu, Nepal

2024

- Conducted **20+ lectures and mentoring sessions** on AI and Machine Learning for undergraduate students.

KEY PROJECTS

Cross-SDK Propagation Agent

 [GitHub](#)

- Designed and fine-tuned **Qwen3-32B** using a two-stage **QLoRA** pipeline on **AWS SageMaker** across 6 repositories (engine + Python/Java/Go SDKs, test contracts, docs); trained on 54,000 generated Q&A pairs to reach **75% closed-book domain reasoning accuracy**, a 45-point gain over the base model (30%).
- Built a **RAG** backbone with an **AST-based chunking pipeline** (1,600+ files → 11,449 function-boundary chunks), **Qdrant** vector database, and **BGE-M3** 1024-dim embeddings; achieved **85% cross-SDK bug-fix propagation** across Python/Java/Go in a single inference, outperforming Claude 3.5 by 25 points.

Bank Note OCR: Vector Similarity and Computer Vision

 [GitHub](#)

- Built a currency detection pipeline using computer vision and deep learning to accurately classify Nepali banknote denominations in real time.
- Applied vector similarity techniques for robust denomination matching under varied lighting and image quality conditions.

Recruit Nepal: AI-Powered Job Matching & ATS Platform

 [GitHub](#)

- Built an end-to-end ATS pipeline with Tesseract OCR fallback, spaCy NER for structured JSON extraction, and Sentence Transformer embeddings.
- Applied cosine similarity for Top-N candidate shortlisting followed by **LLM-based contextual ATS scoring** and automated candidate ranking.

Shishu Care: Infant Cry Classification System

 [GitHub](#)

- Designed a **CNN-BiLSTM with attention** architecture to classify infant cries (hunger, pain, burping, sleep) from mel spectrograms with near real-time inference.
- Performed full audio preprocessing: noise handling, waveform normalisation, and mel spectrogram extraction from raw .wav files.

RAG Chatbot with pgvector

 [GitHub](#)

- Built a production-ready RAG chatbot using **FastAPI + PostgreSQL (pgvector) + Gemini LLM API** with Sentence Transformer embeddings for cosine-similarity nearest-neighbour retrieval.
- Containerised with **Docker-Compose**; interactive Streamlit frontend for querying.

Protein Function Prediction- HealthCare (CAFA-6)

 [GitHub](#)

- Predicted Gene Ontology (GO) labels from amino acid sequences using deep learning, with bioinformatics preprocessing, custom encoding, and class-weighted training.

CERTIFICATIONS

- **Python Programming ,NumPY ,Structured Query Language** – Programiz Pro, 2024
- **Building Safe and Responsible Generative AI Applications with Amazon Bedrock Guardrails** – Amazon Web Services
- **Fundamental of ML and AI** – Amazon Web Services
- **Data Engineering for GEN-AI Application** – Amazon Web Services

PROFESSIONAL REFERENCES

- **Manish Singh** –Senior Cloud Engineer,Amazon
manish@samnilabs.com, +1 (469) 601-1056,
- **Pukar Upreti** – Bunnings, Software Engineer,Australia
pukarupreti3@gmail.com,0467334451

ADDITIONAL QUALIFICATIONS

- **Hackathon Achievement:** Secured **2nd place** at the *Code for Change* event; also participated in Kings College Hackademia hackathon focused on AI and ML, demonstrating strong applied problem-solving skills under competitive conditions, attended UNESCO global hackathon with an idea to detect fake Video based on the LLM and AI stack and attended junction X hackathon based on the finland Embassy and SumX .